

## **New thyroid scintigraphy datasets: Construction and benchmark assessment in diagnosis of residual thyroid tissue**

Lai Phu Minh<sup>1</sup>, Nguyen Chi Thanh<sup>2</sup>, Phung Nhu Hai<sup>2\*</sup>, Dang Nam Thang<sup>3</sup>,  
Nguyen Thanh Trung<sup>3</sup>, Chu Minh Duc<sup>4</sup>, Nguyen Thai Ha<sup>1</sup>, Nguyen Duc Thuan<sup>1</sup>

<sup>1</sup>School of Electrical and Electronic Engineering, Hanoi University of Science and Technology;

<sup>2</sup>Institute of Information Technology, Academy of Military Science and Technology;

<sup>3</sup>Medical Equipment of Department, 108 Medical Central Hospital;

<sup>4</sup>Nuclear Medicine of Department, 108 Medical Central Hospital.

\*Corresponding author: hainda59@gmail.com

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### **ABSTRACT**

*Thyroid scintigraphy, a type of single photon emission computed tomography (SPECT) imaging technique that uses radioactive isotopes to capture images of the thyroid gland, helps detect thyroid abnormalities and diagnosing thyroid cancer. A promising research direction for machine learning applications to assist in diagnosis. Most algorithms for detecting and predicting uptake in the thyroid region rely on proprietary or published datasets with unspecified information. This makes comparing the performance among different methods and developing solutions for various problems challenging. To address this issue, we have constructed two standardized datasets of thyroid scintigraphy images for identifying and quantifying the depth. The purpose of designing the models is to establish a benchmark assessment for developing CADx models on the datasets in the future.*

**Keywords:** SPECT image; Thyroid scintigraphy; Computer-Aided Diagnosis; Residual thyroid tissue; Transfer learning.

### **1. INTRODUCTION**

CADx (Computer-Aided Diagnosis) systems are now widely used in medical diagnosis to alleviate the burden on doctors and to minimize errors. These systems are becoming increasingly [11, 12] popular as they enhance diagnostic precision and eliminate unwanted confusion in the diagnostic process. While CADx systems are employed to improve diagnostic accuracy for various illnesses, there is potential for their application in thyroid cancer scintigraphy, a relatively new imaging area.

Regarding thyroid cancer, standard treatment options include surgery, <sup>131</sup>I treatment, and TSH inhibitor therapy. For patients with thyroid issues, a whole-body scintigraphy scan is usually conducted 4-6 weeks following a complete thyroidectomy before removing thyroid tissue with <sup>131</sup>I [10]. At this point, thyroid scintigraphy is performed to determine the extent and distribution of the remaining thyroid tissue post-surgery and determine the appropriate treatment course. As a result, a vast dataset of thyroid scan images is generated and is available throughout the patient's treatment journey. This collection of thyroid scintigraphy images can be effectively utilized for CADx models to aid specialized physicians.

Although CADx has shown promising research outcomes, it has not been authorized for medical applications yet. One of the reasons behind this is the lack of a standardized database to assess diagnostic assistance systems. Additionally, a CADx model with high precision on a limited dataset cannot be a reliable foundation for ensuring the safety of utilizing such models in real-world scenarios.

Only one thyroid scintigraphy dataset, including SPECT images of 446 patients, has been mentioned in scientific literature. However, it has only been used for Yinxiang Guo [10] and

colleagues' research group, which focuses on the classification and diagnosis of residual thyroid tissue using fine-tuned CNN-based models. Regrettably, sharing this dataset with other researchers has been limited, creating challenges in verifying and utilizing research outcomes related to thyroid scan analysis. The use of proprietary datasets has contributed to slowing down research and the construction of CADx models. As a result, research groups wishing to enter the field often have to collect data from scratch or collaborate with institutions in nuclear medicine, which can be time-consuming.

To address this issue, this article provides the following contributions:

- Contributed two standardized datasets of considerable magnitude by drawing from the most credible data sources in Vietnam and the surrounding areas. These extensive datasets can aid both domestic and international research teams in the development of CADx works in the future for analyzing thyroid scans.

- Provided impartial evaluations of the datasets using state-of-the-art machine learning models, which can be a benchmark for developing CADx models and machine learning-based architectures.

## 2. RELATED WORKS

The accuracy and success of studies in computer vision depend significantly on the quality and accuracy of data. Research has shown that open, standardized datasets are essential for the success of machine learning classification techniques such as deep learning. Despite of many studies on thyroid scintigraphy, there is still a gap in detecting remaining thyroid tissue after surgery. The current datasets available for studying thyroid tissue through thyroid scintigraphy are small, limited to a narrow range of facilities, and have not been publicly released.

**Table 1.** Statistical table of CT and thyroid scan data sets.

| Statistical table of performance of machine learning methods to detect suspected lesions |         |                    |         |         |                 |          |
|------------------------------------------------------------------------------------------|---------|--------------------|---------|---------|-----------------|----------|
| Author                                                                                   | Samples | Publicly available | Acc (%) | Sen (%) | Pseudo positive | Modality |
| Karssemeijer et al.[1]                                                                   | 50      | Yes (MIAS)         | NA      | 90%     | 1               | CT       |
| Mudigonda et al.[2]                                                                      | 56      | Yes (MIAS)         | NA      | 81      | 2.2             | CT       |
| Liu et al.[3]                                                                            | 38      | Yes (MIAS)         | NA      | 90      | 1               | CT       |
| Li et al.[4]                                                                             | 94      | No                 | NA      | 91      | 3.21            | CT       |
| Baum et al.[5]                                                                           | 63      | No                 | NA      | 89      | 0.61            | CT       |
| Kim et al.[6]                                                                            | 83      | No                 | NA      | 96      | 0.2             | CT       |
| Yang et al.[7]                                                                           | 203     | No                 | 96.1    | 95-98   | 1.8             | CT       |
| The et al.[8]                                                                            | 123     | No                 | NA      | 94      | 2.3 per case    | CT       |
| Sadaf et al.[9]                                                                          | 127     | No                 | NA      | 91      | NA              | CT       |
| Yinxian Guo et al.[10]                                                                   | 446     | No                 | 99.96   | 99.6    | NA              | SPECT    |

In the medical field, computed tomography (CT) generates anatomical images, while single photon emission tomography (SPECT) produces metabolic images. Although both are radiographs, they use similar reconstruction algorithms such as filtered back project (FBP) or iterative algorithm. The structure and presentation of the CT image dataset and the SPECT image dataset are quite similar. However, the CT dataset is more comprehensive and accessible than the SPECT dataset. Several good datasets are available, but most are limited in size and are mainly focused on CT scans, such as Digital Database for Screening Mammography (DDSM),

Mammographic Imaging Analysis Society (MIAS), and Image Retrieval in Medical Applications (IRMA). While these datasets are publicly available, they are limited in size. Moreover, some larger datasets exist but are not publicly available. Therefore, while there are ten datasets mentioned in scientific articles for CT image data, there is only one dataset for SPECT data on thyroid, which is the dataset of Yinxiang Guo et al. listed in table 1. However, this dataset is also not widely shared.

The acquisition, selection, and dissemination of imaging data in nuclear medicine, specifically for SPECT in the diagnosis of thyroid cancer, remain unaddressed. Difficulties in data collection and processing, limited technical abilities of hospital teams, and restricted data access have made it challenging to develop a standard SPECT dataset. Several studies in Nuclear Medicine have demonstrated that each population group presents unique pathological characteristics on SPECT images. As a result, it is necessary to create a separate database for Vietnamese individuals to support SPECT diagnosis research in this population. Hence, our research obtained all patient data from Vietnamese individuals.

### 3. DATA CONSTRUCTION

#### 3.1. Data collection

The research team's dataset was acquired from the Department of Nuclear Medicine at 108 Central Military Hospital, consisting of 1,777 scans taken between January 2020 and December 2021 to remove thyroid tissue after treatment. The data was obtained with consultation from multiple doctors and used to treat patients. The study was approved by the Department of Nuclear Medicine at Central Hospital 108. For each case, SPECT images were taken before and after the treatment to capture the entire thyroid tissue. In this paper, we take part of the above data set including 559 images for the experiment. Table 2 shows the characteristics of the research objects.

*Table 2. Characteristics of study subjects.*

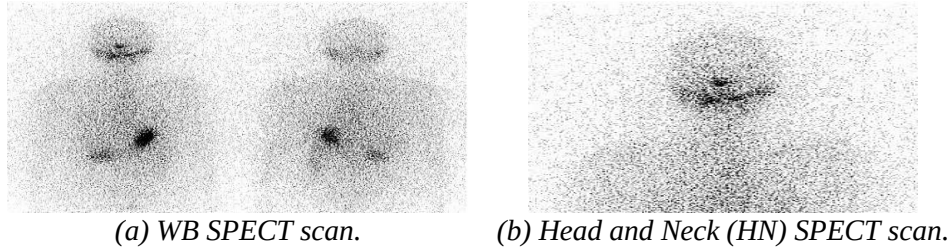
| Parameter                          | Value     |
|------------------------------------|-----------|
| Mean age (years)                   | 39 ± 5,5  |
| Range                              | 21 ~ 59   |
| <b>Patient gender</b>              |           |
| Male                               | 421 (75%) |
| Female                             | 138 (25%) |
| Male/Female Ratio                  | 1/4       |
| <b>Papillary thyroid carcinoma</b> |           |
| Medullary thyroid carcinoma        | 15 (3%)   |
| Undifferentiated thyroid carcinoma | 544 (97%) |

The data was collected from patients who received their initial treatment for the removal of thyroid tissue. Following the treatment, the patients were administered an oral dose of 5 mci and underwent a thyroid scan 36-48 hours later. The dataset contains information marked by doctors to identify residual thyroid tissue areas detected after scintigraphy.

#### 3.2. Data labeling

The patient image data set is labeled by determining whether radioiodine uptake or radioiodine non-uptake on spect images. This process is based on the analysis of the doctor's diagnostic conclusions for each scan. This assessment is rigorously checked during the panel consultation on a case-by-case basis to ensure the correctness of the dataset. From that data, the machine learning process will use and produce images with or radioiodine non-uptake of the patient after total thyroidectomy.

### 3.3. Dataset construction



**Figure 2.** Sample images from two datasets.

We constructed two datasets from the collected data: (i) Full Body Dataset, which combines front and back SPECT images, and (ii) Neck Region Image Dataset, which is obtained by extracting images of the neck from the front SPECT images. These datasets are further divided into three subsets: Train, Validation, and Test sets. The exact numbers for each subset are provided in table 3.

**Table 3.** Description of sub-datasets.

|              | Samples     | Radioiodine Uptake | Radioiodine Non-Uptake |
|--------------|-------------|--------------------|------------------------|
| Train        | 359 (64,2%) | 199                | 160                    |
| Validation   | 88 (15,7%)  | 49                 | 39                     |
| Test         | 112 (20,1%) | 62                 | 50                     |
| <b>Total</b> | <b>559</b>  | <b>310</b>         | <b>249</b>             |

## 4. BASELINE RESULTS

### 4.1. Training method

We applied transfer learning (TL) [11] to test our datasets. TL is a deep learning technique that has been successfully used in medical image analysis, such as bone X-rays [12], chest X-rays [13], cardiac SPECT [14], and brain MRI [15]. TL helps to address the problem of data scarcity, offering better initial accuracy, faster convergence speed, and higher asymptotic accuracy than traditional deep learning methods of training a model from scratch.

We utilized transfer learning techniques to train several modern convolutional network models, including VGG [16], Inception-v3 [17], Resnet [18], Xception [19], MobileNet [20], NASNetMobile [21], and EfficientNet [22]. These models are widely used for image classification tasks in the field of medical imaging, specifically as follows:

- **VGG:** Proposed by the University of Oxford in 2014, VGG is a deep convolutional neural network architecture that uses consecutive convolutional layers with small 3x3 kernel size to replace each other, reducing the number of parameters and increasing accuracy.
- **Inception-v3:** Released in 2015, Inception-v3 is an improved version of Inception-v1 and Inception-v2. It addresses the computational depth and cost issues of previous versions and solves the bottleneck problem, enhancing the learning ability of the model.
- **ResNet:** Proposed by Microsoft in 2015, ResNet is one of the most famous deep convolutional neural network architectures in computer vision. It introduces short connections that transfer information directly from one layer to a non-contiguous layer, helping to avoid the issue of Vanishing Gradient during backpropagation and improving model convergence.
- **Xception:** In 2016, Google proposed the Xception model, an improved version of the Inception architecture. Xception employs the Separable Convolution method, which serves as an alternative to the Inception module architecture, to decrease the number of parameters in the model while maintaining accuracy.

- *MobileNet* is a model designed for mobile devices with limited computational resources. It employs Depthwise Separable Convolution, a special convolution architecture that reduces the number of parameters and optimizes computation. MobileNet-v2, introduced in 2018, further enhances the model's accuracy and performance using Inverted Residuals and Linear Bottlenecks methods.

- *NASNetMobile*, on the other hand, is a convolutional neural network model created by Google Brain using the Neural Architecture Search (NAS) network structure optimization method. Despite its complex architecture, NASNetMobile has a small size and high efficiency, making it a popular choice for computer vision applications on mobile devices.

- *EfficientNet* is a family of convolutional neural network architectures that have been optimized according to the model's input size, width, and depth. This results in a highly efficient model that has achieved excellent performance on many image classification datasets and is considered one of the most efficient computer vision models available today.

#### 4.2. Evaluation metrics

We used the following four metrics to evaluate the performance of the models:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (1)$$

$$Recall = \frac{TP}{TP + FN} \quad (2)$$

$$TNR = \frac{TN}{TN + FP} \quad (3)$$

$$F1 - score = \frac{2TP}{2TP + FN + FP} \quad (4)$$

where, TP (True Positive) represents the number of correctly identified cases of radioiodine uptake; TN (True Negative) represents the number of correctly identified cases of radioiodine non-uptake; FP (False Positive) is the number of radioiodine uptake cases predicted as radioiodine non-uptake; and FN (False Negative) is the number of radioiodine non-uptake cases predicted as radioiodine uptake. A high TNR value indicates fewer misdiagnoses for patients without the disease, while a high Recall value indicates a reduced likelihood of missed cases. The F1-score represents the model's ability to make accurate diagnoses with minimal misdiagnosis for patients without the disease and detect all cases of the disease.

#### 4.3. Results and discussion

The performance of various models on the Test datasets is presented in tables 4 and 5. It is observed that the majority of the models exhibit better performance in all indices on the neck image dataset than on the full-body image dataset. This can be attributed to the fact that the neck images focus only on the thyroid region, where the radioiodine uptake difference between healthy and diseased cases is more apparent. On the other hand, the full-body images contain more information, and the abdominal organs also show an increase in radioiodine uptake similar to the cases in the neck region, making it more challenging for the models to differentiate accurately. This difference necessitates a larger training dataset.

Regarding the results on the full-body SPECT dataset shown in table 3, the best results for each indicator are highlighted in bold. Among all the models tested, Xception performs the best, with the highest scores in three out of four indicators: Accuracy (0.893), Recall (0.952), and F1-score (0.908). However, Xception has a relatively low TNR, indicating that the model is inclined towards predicting an increase in radioiodine uptake label. This can be attributed to the fact that the number of radioiodine uptake labels in the training dataset is 1.24 times higher than the number of radioiodine non-uptake labels. Furthermore, the limited amount of image data is also a

contributing factor to this difference. The remaining models also display this bias, albeit to a lesser extent. Meanwhile, EfficientNetB7 is the only model with a higher TNR than the Recall index, indicating that the label discrepancies influence it less in the full-body training dataset.

**Table 4.** Result of different models on the test set of full-body SPECT images.

| Model          | Accuracy     | Recall       | TNR         | F1-score     |
|----------------|--------------|--------------|-------------|--------------|
| VGG16          | 0.884        | 0.919        | 0.840       | 0.898        |
| InceptionV3    | <b>0.893</b> | 0.919        | 0.860       | 0.905        |
| ResNet152V2    | <b>0.893</b> | 0.935        | 0.840       | 0.906        |
| Xception       | <b>0.893</b> | <b>0.952</b> | 0.820       | <b>0.908</b> |
| MobileNetV2    | <b>0.893</b> | 0.935        | 0.840       | 0.906        |
| NASNetMobile   | 0.875        | 0.887        | 0.860       | 0.887        |
| EfficientNetB0 | 0.866        | 0.903        | 0.820       | 0.882        |
| EfficientNetB7 | <b>0.893</b> | 0.871        | <b>0.92</b> | 0.900        |

Table 5 shows the evaluation results of the models on the SPECT dataset. The Xception and VGG16 models achieved the best performance with the highest scores in three out of four metrics: Accuracy, Recall, and F1-score, reaching 0.955, 0.984, and 0.961 respectively, and a high TNR value of 0.920. Moreover, the smaller-sized NASNetMobile model also performed well, with a Recall index equal to that of Xception and VGG16 at 0.984, and other indicators showing positive results as well.

**Table 5.** Results of different models on the SPECT images test set of the neck region.

| Model          | Accuracy     | Recall       | TNR          | F1-score     |
|----------------|--------------|--------------|--------------|--------------|
| VGG16          | <b>0.955</b> | <b>0.984</b> | 0.920        | <b>0.961</b> |
| InceptionV3    | 0.911        | 0.887        | <b>0.940</b> | 0.917        |
| ResNet152V2    | 0.911        | 0.903        | 0.920        | 0.918        |
| Xception       | <b>0.955</b> | <b>0.984</b> | 0.920        | <b>0.961</b> |
| MobileNetV2    | 0.920        | 0.919        | 0.920        | 0.927        |
| NASNetMobile   | 0.946        | <b>0.984</b> | 0.900        | 0.953        |
| EfficientNetB0 | 0.911        | 0.952        | 0.860        | 0.922        |
| EfficientNetB7 | 0.884        | 0.903        | 0.860        | 0.896        |

The experiment conducted above has demonstrated the possibilities for exploring and creating deep learning models that can aid in diagnosing and detecting thyroid cancer post-treatment in the future. To accomplish this, there is a need to address the issue of data imbalance and enhance model performance using models such as Xception and EfficientNetB7 for the full-body SPECT dataset and Xception, VGG16, and NASNetMobile for the neck region dataset.

## 5. CONCLUSIONS

The paper describes two datasets which are both trustworthy and accessible for developing CADx models to identify cervical scintillation via thyroid scintigraphy analysis. Additionally, the research community has conducted experiments on this dataset using image classification models.

The datasets are the first of its kind for thyroid scintigraphy, providing ample, precise, dependable, and extensively accessible information. We aspire that further research into machine learning-based diagnostic assistance will be conducted using the datasets. The experiments performed may serve as an initial foundation for future CADx modeling research.

Moving forward, we plan to gather more data to expand the sample size and enhance models that can aid in identifying radioactive areas in the thyroid and detecting metastasis in other regions of the body. We also aim to collect additional clinical indicators that can assist in determining the appropriate treatment plan for the patient.

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### TÓM TẮT

#### **Đánh giá cấu trúc và điểm chuẩn trong chẩn đoán mô tuyến giáp còn sót lại trên bộ dữ liệu xạ hình tuyến giáp**

Xạ hình tuyến giáp, một loại kỹ thuật chụp cắt lớp vi tính phát xạ photon đơn (SPECT) sử dụng đồng vị phóng xạ để chụp ảnh tuyến giáp, giúp phát hiện các bất thường về tuyến giáp và chẩn đoán ung thư tuyến giáp. Một hướng nghiên cứu triển vọng ứng dụng học máy hỗ trợ trong chẩn đoán. Hầu hết các thuật toán để phát hiện và dự đoán sự hấp thu ở vùng tuyến giáp đều dựa trên các bộ dữ liệu độc quyền hoặc đã công bố với thông tin không xác định. Điều này làm cho việc so sánh hiệu suất giữa các phương pháp khác nhau và phát triển các giải pháp cho các vấn đề trở nên khó khăn. Để giải quyết vấn đề này, chúng tôi đã xây dựng hai bộ dữ liệu tiêu chuẩn về hình ảnh xạ hình tuyến giáp để xác định và định lượng độ sâu vùng quan tâm. Mục đích của việc thiết kế các mô hình là để thiết lập đánh giá chuẩn cho việc phát triển các mô hình CADx trên bộ dữ liệu trong tương lai.

**Từ khóa:** Chụp cắt lớp phát xạ đơn photon; Xạ hình tuyến giáp; Hỗ trợ chẩn đoán bằng máy tính CADx.